ANTIEMETICS



SPECIFIC LEARNING OBJECTIVES

At the end of the session the student must be able to

1. Analyze the mechanism of emesis and relate the sites of action of various antiemetic drugs in blocking the emetic pathway.
2. Analyze and differentiate the mechanisms of action, therapeutic uses, and adverse effects of prokinetic agents, serotonergic antagonists, and antihistamines.
3. Evaluate and select appropriate drugs for different types and causes of vomiting based on their mechanisms of action.

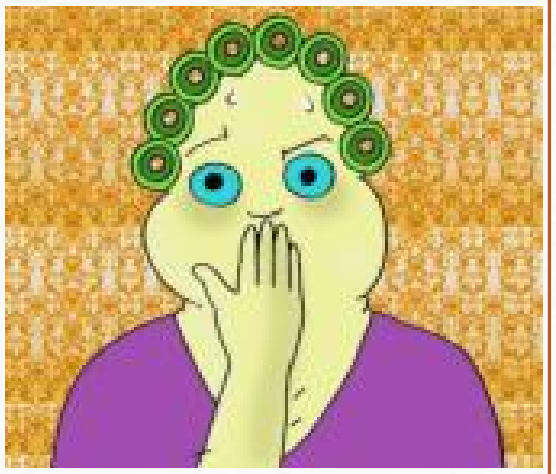
- Nausea- a subjective, unpleasant feeling of the need to vomit. It may be accompanied by tachycardia, diaphoresis and salivation

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- Retching- is the reverse movement of the stomach and esophagus without vomiting

- Vomiting- forceful expulsion of gastric contents due to powerful and sustained contraction of the muscles of the abdomen and thorax.

VOMITING



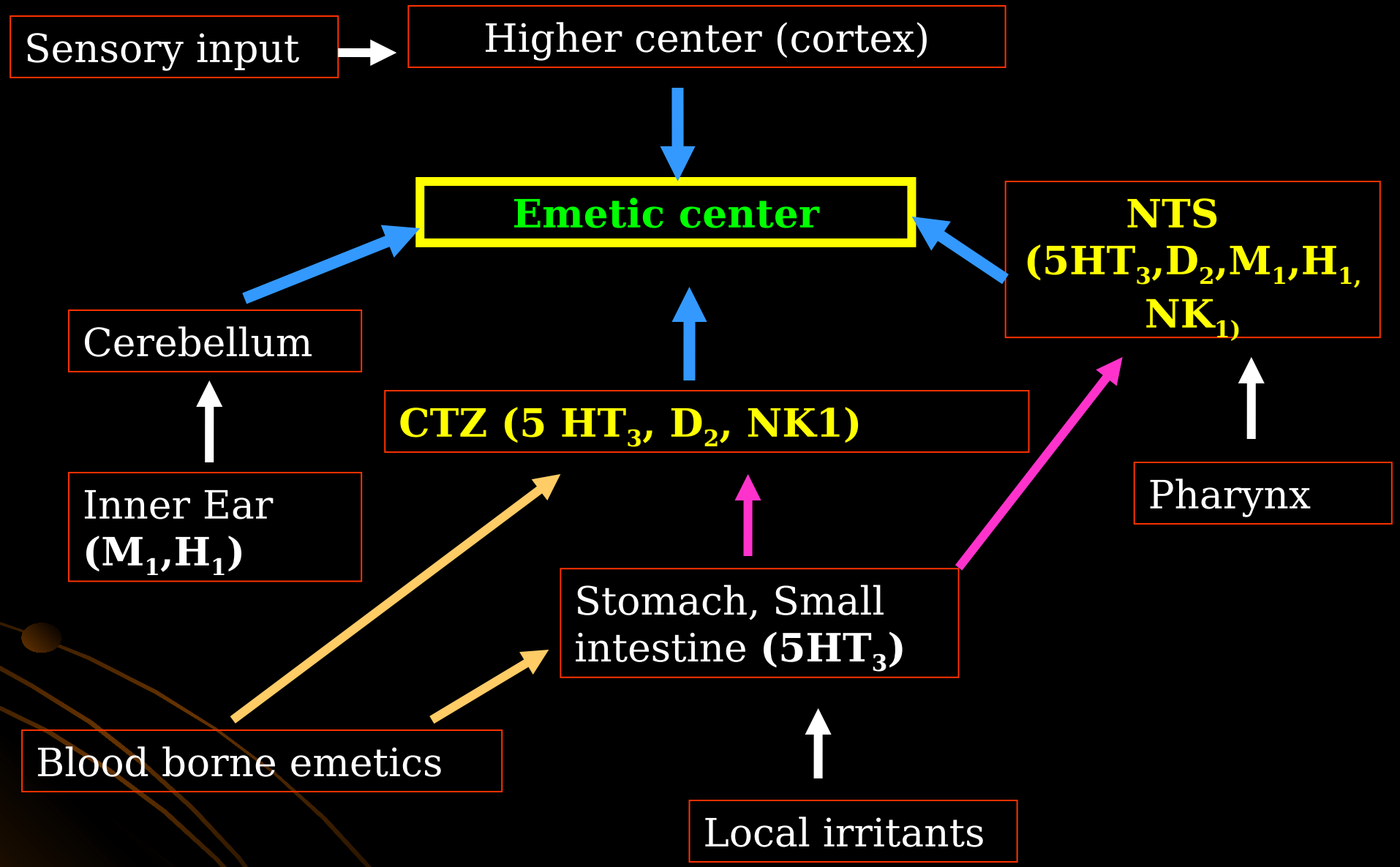
- Vomiting is a complex process that consists of :

- **PRE-EJECTION PHASE:**
  Gastric relaxation and retro peristalsis.

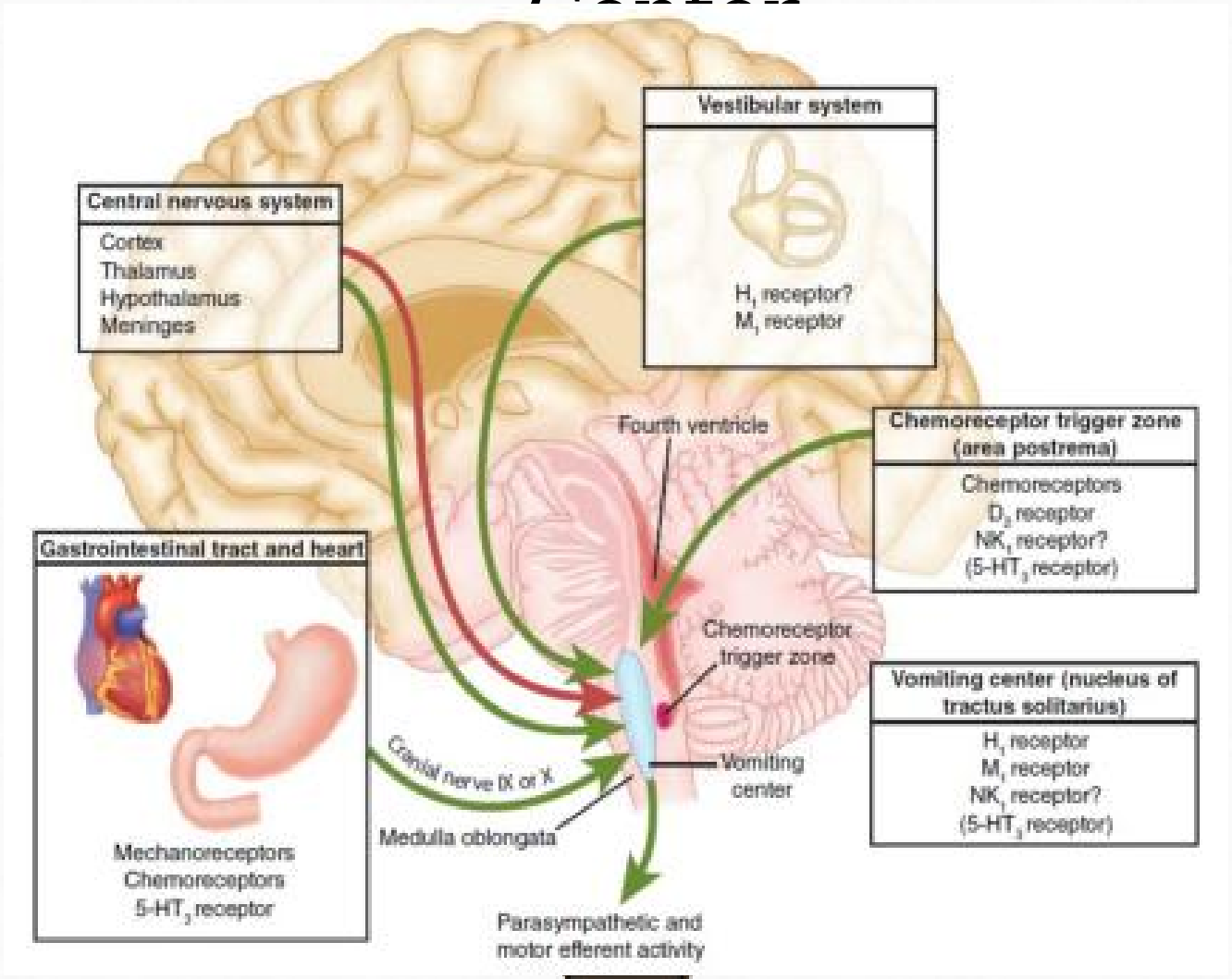
- **RETCHING:**
  Rhythmic action of respiratory muscles preceding vomiting and consist of abdominal & intercoastal muscles and diaphragm against a closed glottis.

- **EJECTION:**
  Intense contraction of abdominal muscles and relaxation of upper oesophageal sphincter.

- Followed by multiple autonomic phenomena:
  Salivation, Shivering, Vasomotor changes


Sensory input
Higher center (cortex)
Emetic center
NTS
(5HT3,D2,M1,H1,
NK1)
Cerebellum
CTZ (5 HT3, D2, NK1)
Inner Ear
(M1,H1)
Pharynx
Stomach, Small
intestine (5HT3)
Blood borne emetics
Local irritants

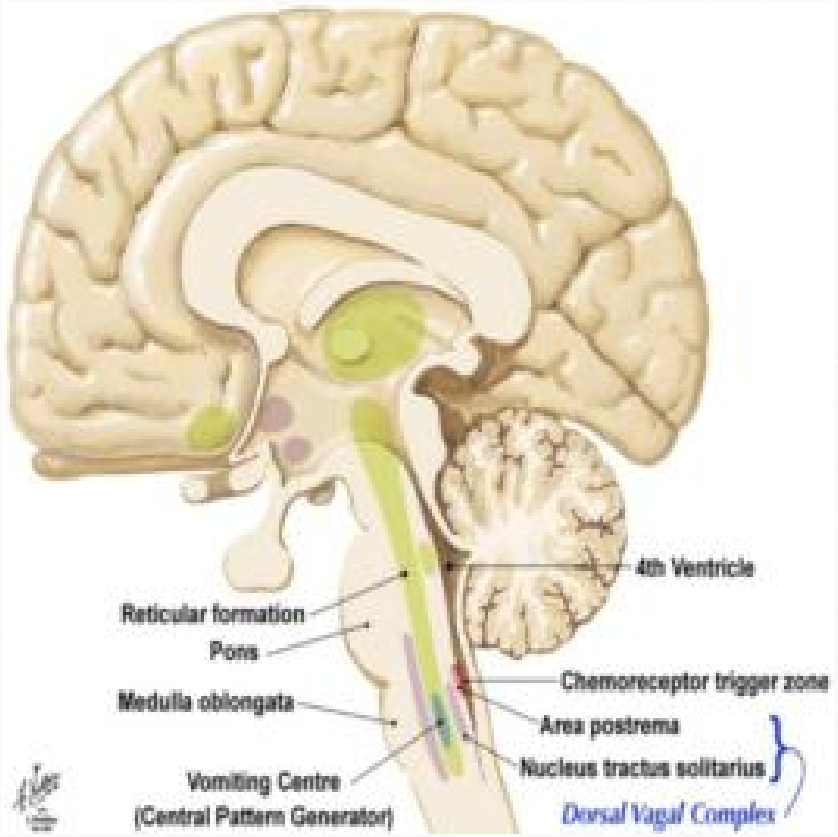
Four Afferent Inputs to Vomiting Center



1. CTZ (Area Postrema)

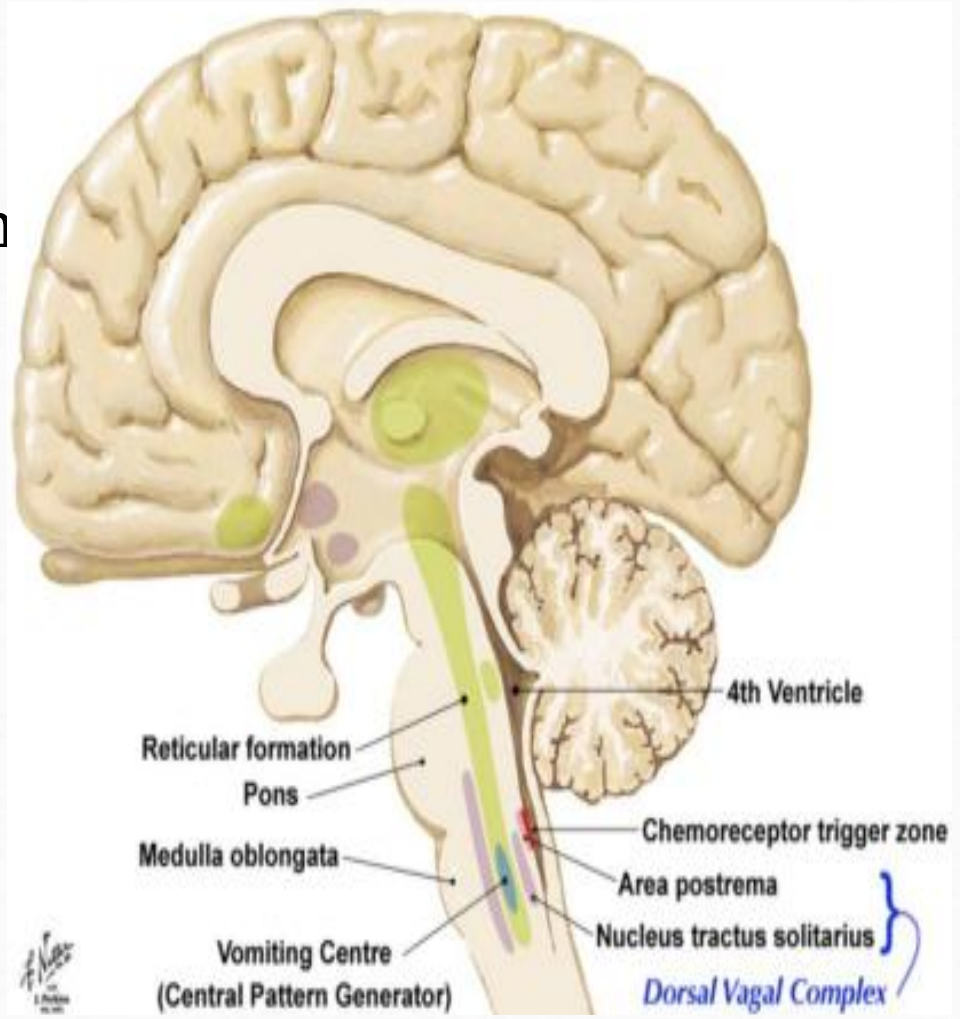
- Located in floor of fourth ventricle, outside the BBB
- Play a role in any vomiting due to blood borne emetics
- Rich in D2, 5-HT-3 and NK receptors

- Antiemetics:
  - Ondansetron (5-HT3 antagonist)
  - Metoclopramide (Dopamine antagonist)
  - Domperidone (Dopamine antagonist)
  - Aprepitant (NK antagonist)



2. Vestibular System

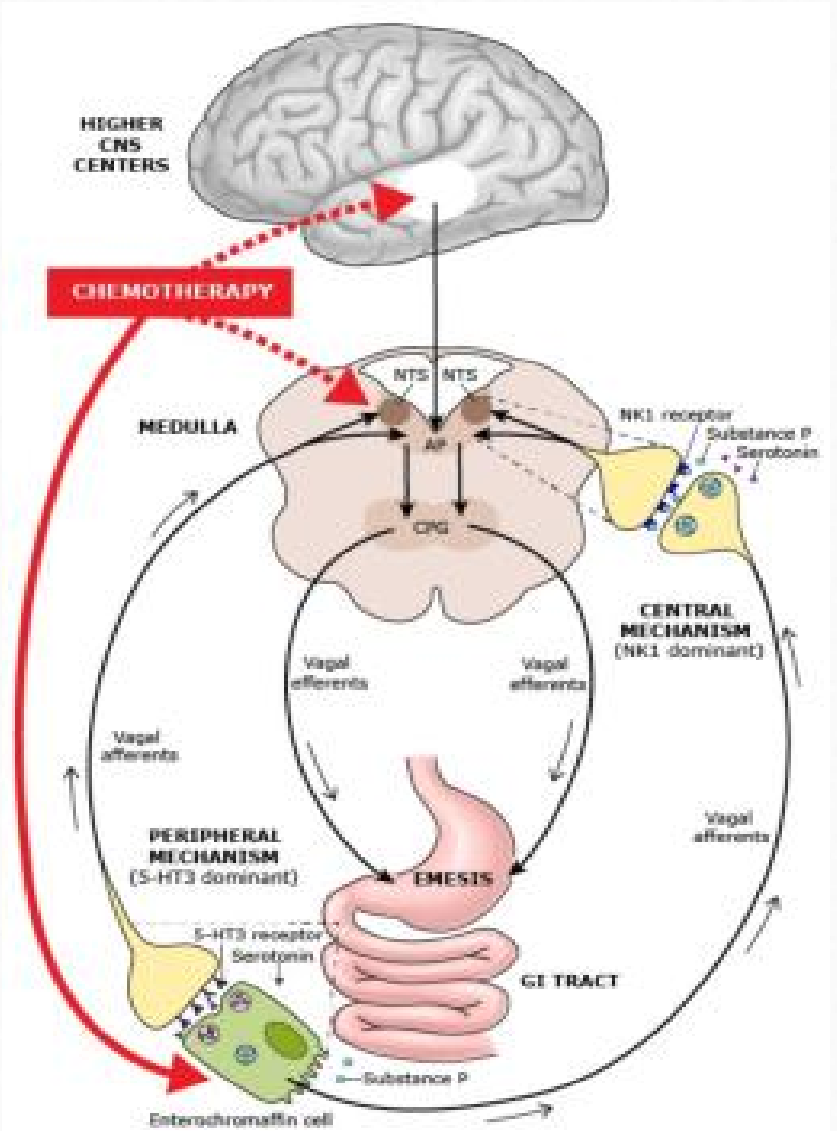
- Play a role in vomiting due to motion sickness
- Mediated through VIII cranial nerve
- Rich in M1 and H1 receptors
- Antiemetics:
  - Anticholinergics : **Scopolamine** (Hyoscine)
  - Antihistamines : Meclizine, Diphenhydramine



3. GIT

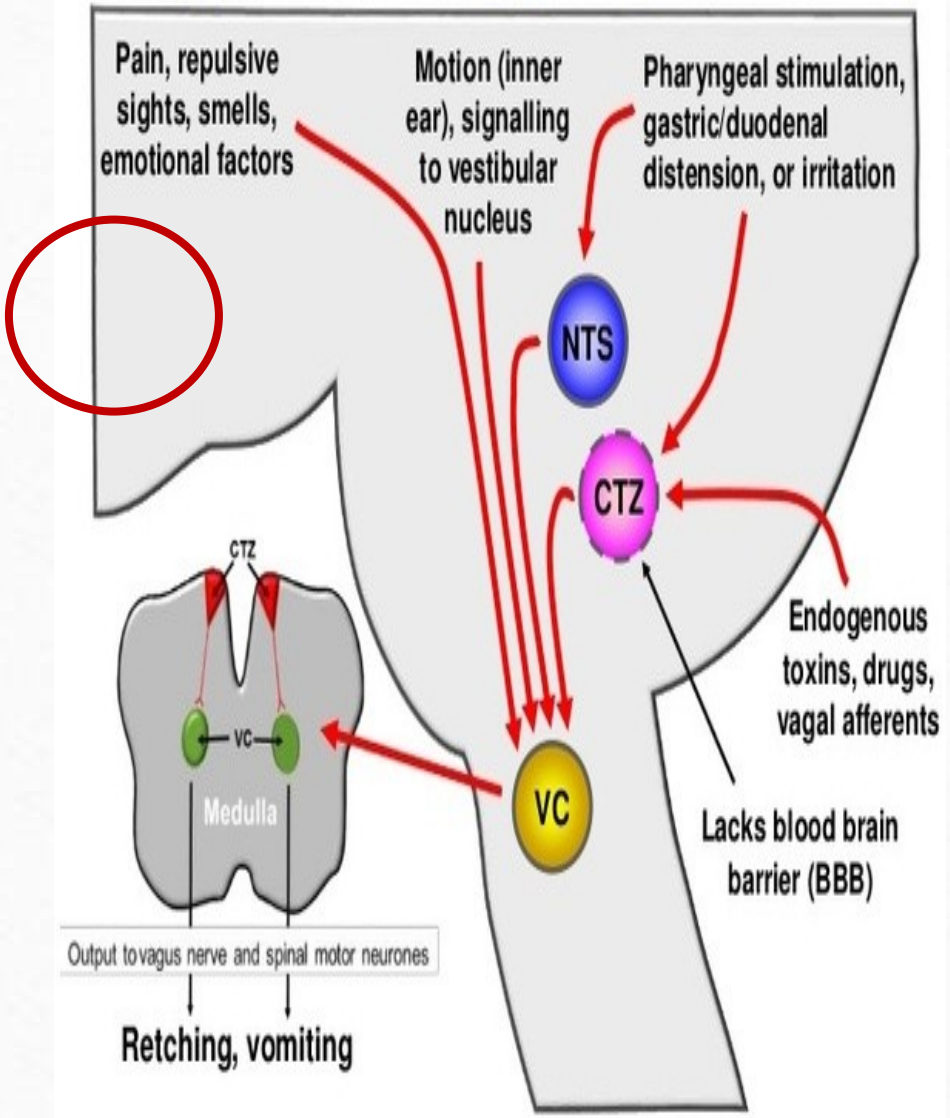
- Play a role in vomiting caused by GI irritation due to:
  - Chemotherapy
  - Radiotherapy
  - Enteritis
- Mediated through Vagus nerve and other spinal nerves
- Rich in 5-HT_3 receptors
- Antiemetics: Ondansetron

Granisetron , Dolasetron, Palonosetron



4. CNS

- Play a role in vomiting due to psychiatric disorders, stress and anxiety
- Anticipatory vomiting prior to chemotherapy
- Antiemetics:
  - Benzodiazepines
  - Dronabinol
  - Phenothiazines: Prochlorperazine and promethazine which reduces vomiting by blocking M and D2 receptors (They also blocks H1: causes sedation)
  - Neuroleptics: Chlorpromazine,



CLASSIFICATION

Anticholinergics:	Hyoscine, Dicyclomine
H1 antihistaminics	Promethazine, Diphenhydramine, Doxylamine, Cyclizine, Meclizine and Cinnarazine
Prokinetic drugs	Metoclopramide, Domperidone, Cisapride, Mosapride and Tegaserod
5HT3 antagonists	Ondansetron, Granisetron, Dolansetron, Palonosetron, Ramosetron, Tropisetron
Neuroleptics (D2 blockers	Chlorpromazine, Prochlorperazine and Haloperidol
NK_1 receptor antagonist	Aprepitant
CANABINOID RECEPTOR AGONIST	Dronabinol, Nabilone
Others	Dexamethasone,Benzodiazepines

Anticholinergics

Hyoscine: Effective drug - motion sickness

- Brief duration of action; produces sedation & other anticholinergic side effects
- Action:

  - Blocks the afferent impulses to vomiting centers

  - Antimuscarinic action

- Available as a transdermal patch

Dicyclomine: Used for prophylaxis of motion sickness & morning sickness

A 43-year-old woman develops nausea during a cruise vacation. She is treated with a transdermal medication. The drug works primarily by inhibiting muscarinic signaling from the vestibular apparatus to the vomiting center.

Which adverse effect is most likely?

A. Diarrhea
B. Bradycardia
C. Excess salivation
D. Dry mouth
E. Parkinsonism

H_1 Antihistaminics

- Primarily used in motion sickness, morning sickness and some other vomiting in lesser extent – also anticholinergic, antihistaminic and antidopaminergic actions
- Promethazine (Phenothiazine), diphenhydramine: 4 – 6 Hours protection
  - Combined with metoclopramide: additive effect plus counters extra pyramidal effects
- Promethazine theoclate (Avomine) – motion sickness
- Doxylamine: Sedative H1 antihistaminic – marketed in combination with Pyridoxine – specifically for morning sickness – duration of action 10 Hours (at bed time) – drowsiness, dry mouth, vertigo
- Meclizine: Long duration of action – sea sickness
- Cinnarizine: anti vertigo action – inhibits Ca++ influx in endolymph

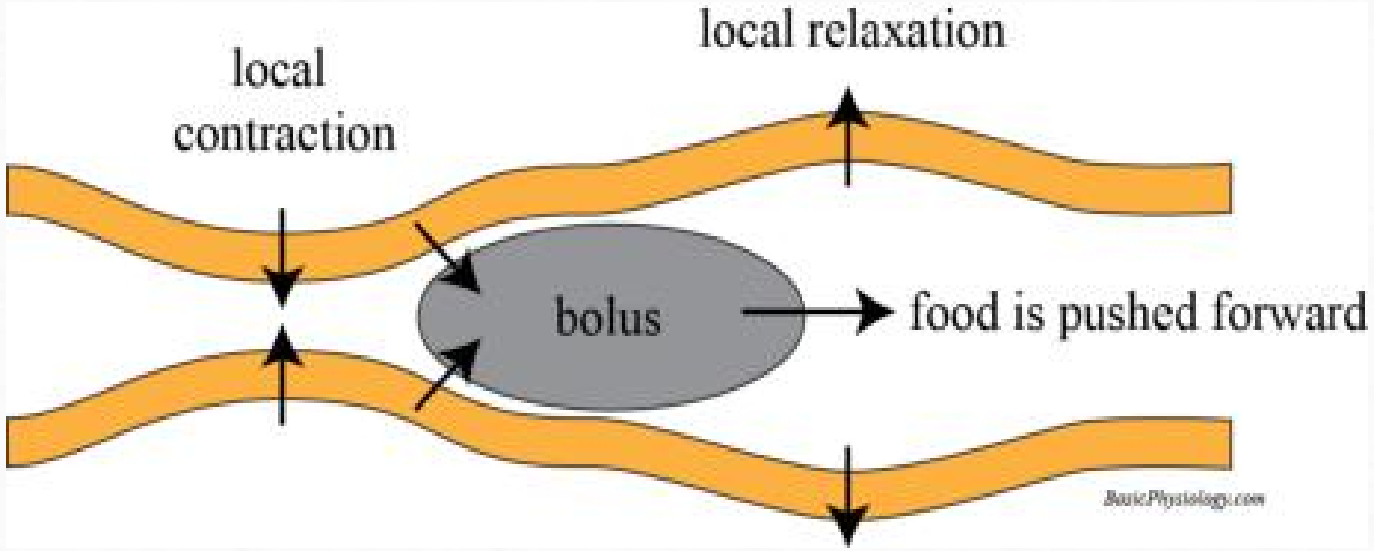


Prokinetic drugs

Drugs which promote gastrointestinal transit and speed gastric emptying

Peristalsis

Peristalsis is a series of reflex responses to a bolus in the lumen results in contraction of the circular muscle on the proximal side of the bolus and relaxation on the distal side.

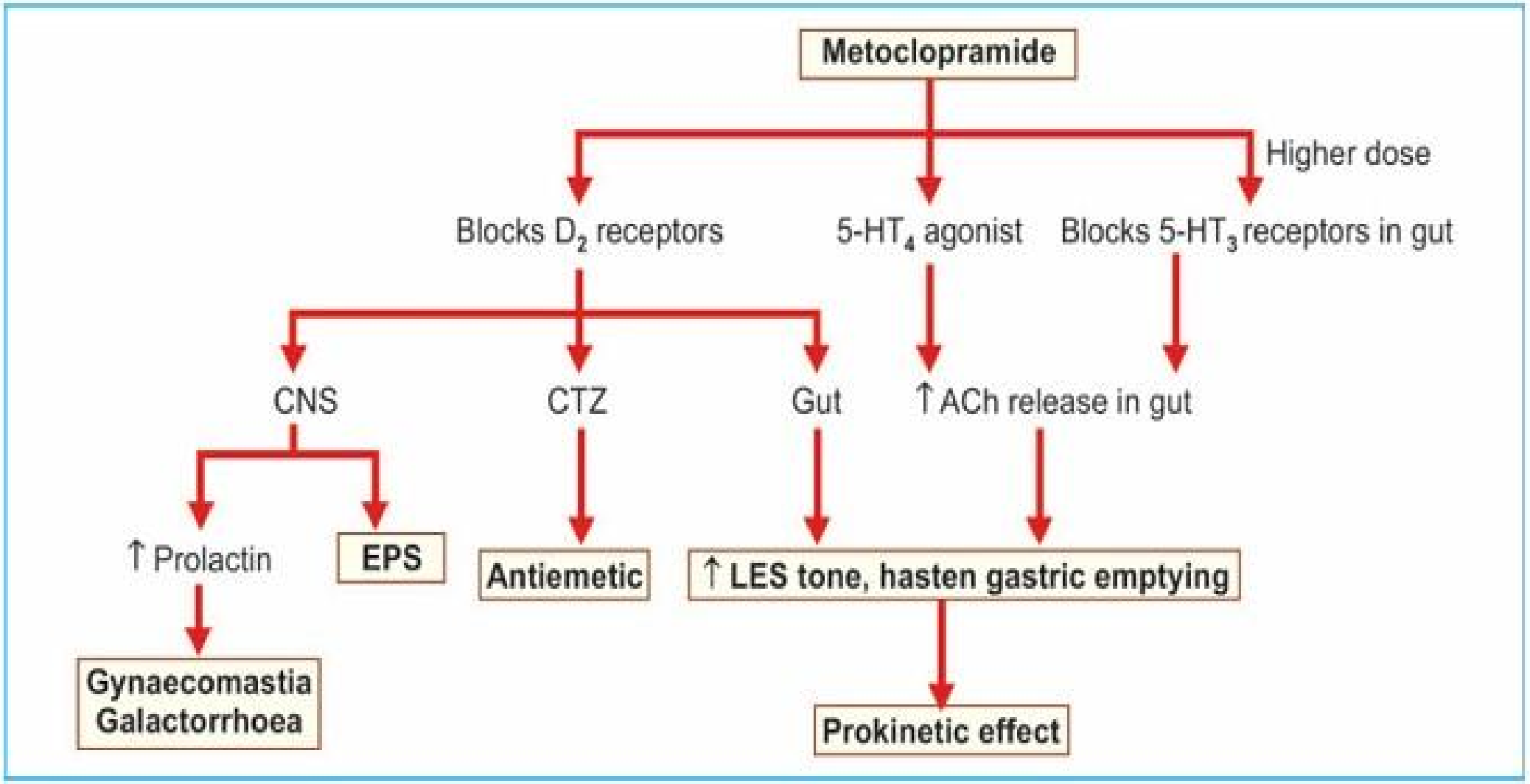


METOCLOPRAMIDE

- Increases gastric peristalsis by relaxing the pylorus & first part of duodenum → speeds gastric emptying
- ↑ Lower oesophageal sphincter tone
- ↑ intestinal peristalsis

MOA

- D2 (receptors) antagonism in CTZ
- 5-HT4 agonism – Enhance ACh release in GIT
- 5-HT3 antagonism in gut

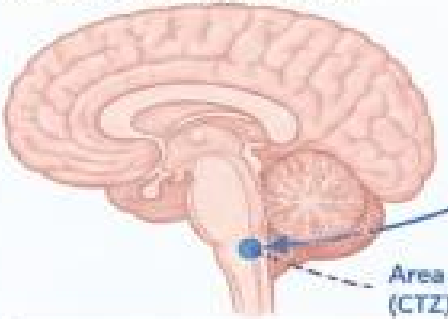

Metoclopramide
Higher dose
Blocks D_2 receptors
5-HT_4 agonist
Blocks 5-HT_3 receptors in gut
CNS
CTZ
Gut
↑ ACh release in gut
↑ Prolactin
EPS
Antiemetic
↑ LES tone, hasten gastric emptying
Gynaecomastia
Galactorrhoea
Prokinetic effect

MECHANISM OF ACTION OF METOCLOPRAMIDE

A dopamine D_2 receptor antagonist and 5-HT_4 receptor agonist

1. CENTRAL ANTIEMETIC ACTION

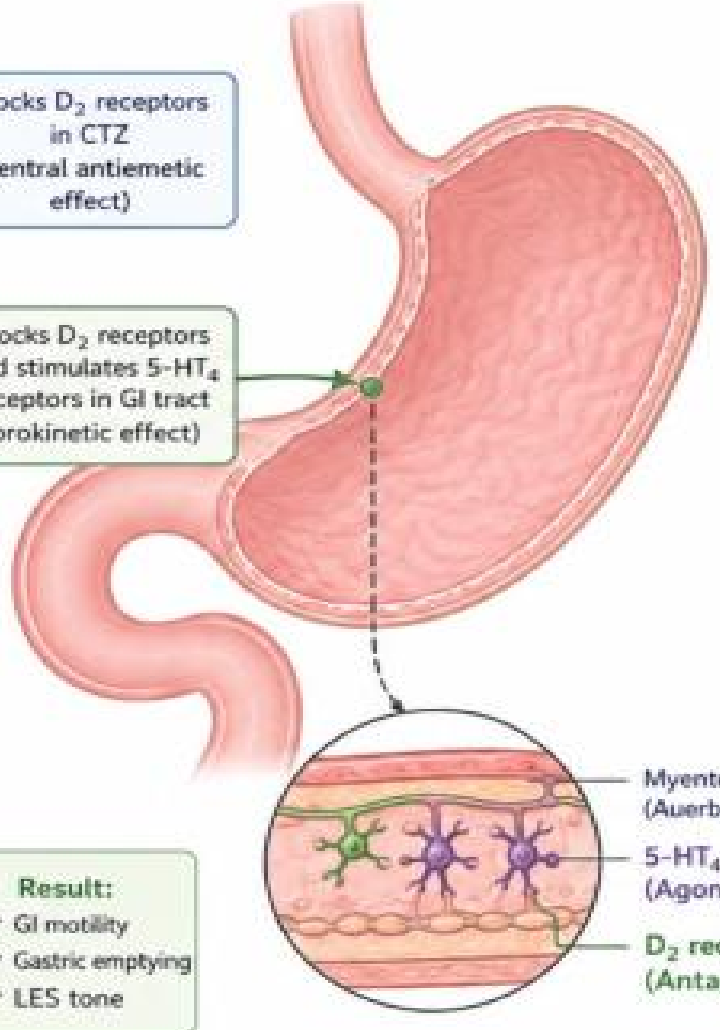
Metoclopramide crosses the blood–brain barrier and acts on the chemoreceptor trigger zone (CTZ) in the area postrema.



MOA: Blocks dopamine D_2 receptors in CTZ → inhibits emetic stimulation → prevents nausea and vomiting.

Blocks D_2 receptors in CTZ (central antiemetic effect)

Blocks D_2 receptors and stimulates 5-HT_4 receptors in GI tract (prokinetic effect)



2. GASTROINTESTINAL PROKINETIC ACTION

Acts on GI tract to enhance motility.

MOA:

- Antagonizes dopamine D_2 receptors in the enteric nervous system → removes dopamine-mediated inhibition → increases acetylcholine release
- Agonizes 5-HT_4 receptors on enteric neurons → enhances acetylcholine release

Result:

- ↑ GI motility
- ↑ Gastric emptying
- ↑ LES tone

SUMMARY OF ACTION

CENTRAL ANTIEMETIC

- Blocks D_2 receptors in CTZ (area postrema)
- Inhibits emetic signals
- Relieves nausea and vomiting

PERIPHERAL PROKINETIC

- Blocks D_2 receptors in enteric nervous system
- Agonizes 5-HT_4 receptors
- Increases acetylcholine release
- Increases GI motility
- Increases gastric emptying
- Increases lower esophageal sphincter tone

OVERALL EFFECT

- Prevents nausea and vomiting
- Enhances gastrointestinal motility
- Useful in gastroparesis, reflux, and drug-induced nausea and vomiting

Drug Class: Dopamine D_2 receptor antagonist, 5-HT_4 receptor agonist | **Uses:** Nausea & vomiting, Gastroparesis, GERD

Pharmacokinetics

- Rapid oral absorption
- Enters brain, crosses placenta & is secreted in milk
- T1/2: 3-6 hrs, Action lasts for 4-6 hrs

Adverse effects:

- Sedation, dizziness, diarrhoea, muscle dystonias, tardive dyskinesia
- Parkinsonism, galactorrhoea, gynaecomastia
- Suckling infant may develop loose motions, dystonias, myoclonus

USES

- Post-operative, drug induced, disease associated, chemotherapy induced, radiation sickness
- Gastroesophageal reflux disease
- As gastrokinetic –
  - when emergency general anesthesia to be given,
  - in acute or chronic gastroparesis,
  - to facilitate duodenal intubation

- Not preferred in motion sickness

A 51-year-old man with diabetic gastroparesis is treated with a drug that increases gastric emptying and blocks dopamine receptors in the chemoreceptor trigger zone (CTZ). After several weeks, he develops muscle rigidity and tremors.

Which drug was most likely prescribed?

A. Ondansetron
B. Aprepitant
C. Metoclopramide
D. Scopolamine
E. Dronabinol

A 64-year-old diabetic man complains of early satiety, bloating, and recurrent vomiting. Gastric emptying studies confirm gastroparesis. He is prescribed a medication that enhances acetylcholine release in the enteric nervous system.

Which additional effect is most likely caused by this medication?

A. Cannabinoid receptor activation
B. Muscarinic receptor blockade
C. Dopamine receptor antagonism in CTZ
D. Histamine receptor antagonism
E. NK_1 receptor blockade

DOMPERIDONE

- Selective peripheral D_2 antagonist
- Does not cross BBB; hence no extra-pyramidal side effects
- Does act on CTZ- not protected by BBB
- Plasma t1/2 = 7.5 hr
- A/E: Dry mouth, loose stools, headache, rashes, galactorrhoea, hyperprolactinemia
- Preferred antiemetic in children & in patients with parkinsonism

MECHANISM OF ACTION OF DOMPERIDONE

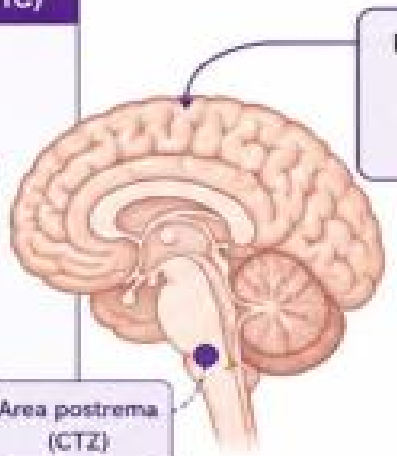
A selective dopamine D_2 receptor antagonist (peripheral action)

1. CENTRAL ACTION (ANTIEMETIC)

Domperidone does NOT cross the blood–brain barrier to a significant extent → acts peripherally.

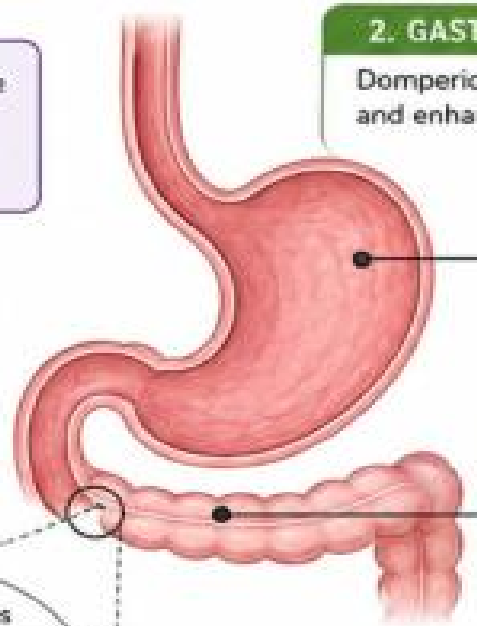
At the Chemoreceptor Trigger Zone (CTZ) in the area postrema:

Blocks D_2 receptors
→ inhibits emetic stimuli
→ prevents nausea and vomiting



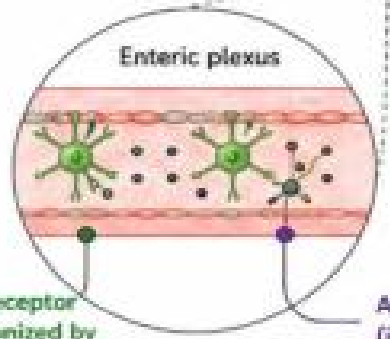
2. GASTROINTESTINAL ACTION (PROKINETIC)

Domperidone antagonizes D_2 receptors in the GI tract and enhances GI motility.



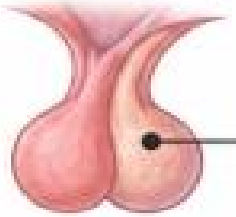
Blocks D_2 receptors in the stomach
→ Inhibits dopamine's inhibitory effect
→ Increases gastric motility
→ Accelerates gastric emptying

Blocks D_2 receptors in enteric nervous system
→ Enhances acetylcholine release
→ Increases GI motility
→ Improves peristalsis



3. EFFECT ON PROLACTIN SECRETION

In the anterior pituitary, domperidone blocks D_2 receptors.



Blocks D_2 receptors on lactotrophs
→ Inhibits dopamine's inhibitory effect
→ Increases prolactin secretion

OVERALL EFFECTS

- Antiemetic: prevents nausea and vomiting
- Prokinetic: increases gastric emptying, GI motility and tone
- Increases lower esophageal sphincter tone
- May increase prolactin levels

DRUG CLASS:

Dopamine D_2 receptor antagonist
(peripheral selective prokinetic & antiemetic)

CLINICAL USES:

- Nausea & vomiting (e.g., diabetic gastroparesis, post-operative, drug-induced)
- Gastroesophageal reflux, functional dyspepsia, delayed gastric emptying

Key Point: Domperidone acts peripherally by blocking D_2 receptors in CTZ and GI tract, enhancing motility and preventing emesis without central side effects.

Pharmacokinetics

- Plasma t1/2 = 3-5 hrs
- Duration of action: 4-12 hrs

- Can be given both orally & IV

Uses

- Vomiting due to drug over dosage, uremia & neurological injuries
- GERD

Mosapride

- 5-HT_4 agonism →promotes Ach release from myenteric motor neurons
- Used only as prokinetic in gastro intestinal motility disorders & GERD

Itopride

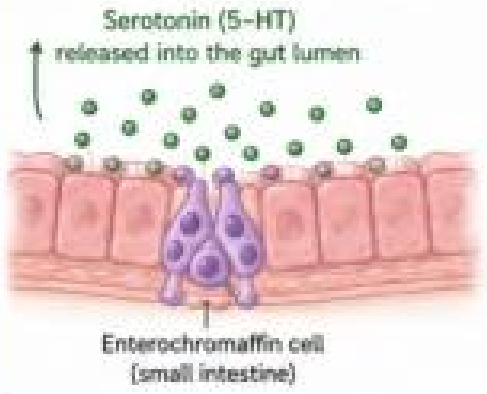
- D2 Blocker & potentiates Ach action
- Side effects: abdominal pain, loose stool & headache

MECHANISM OF ACTION OF 5-HT$_3$ ANTAGONISTS

5-HT$_3$ receptor antagonists block serotonin-mediated signaling involved in nausea and vomiting.

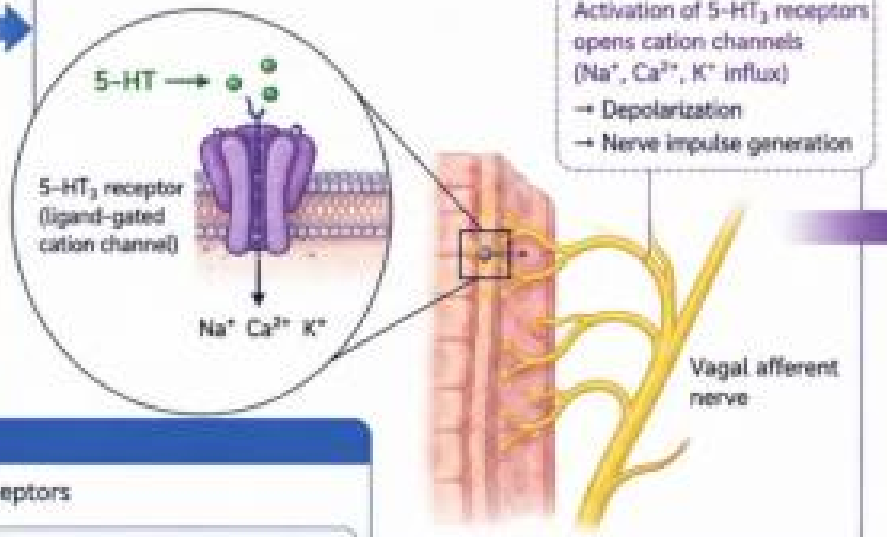
1. RELEASE OF SEROTONIN

Cytotoxic drugs, radiotherapy or surgical stress stimulate enterochromaffin cells in the gut to release serotonin (5-HT).



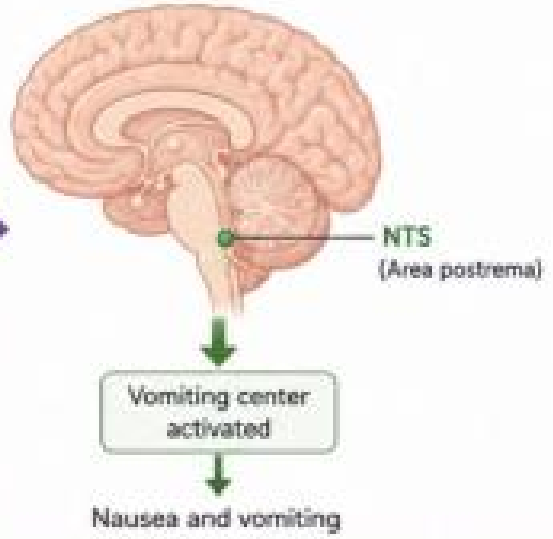
2. ACTIVATION OF 5-HT$_3$ RECEPTORS

Serotonin binds to 5-HT$_3$ receptors on vagal afferent nerve endings in the gut wall.



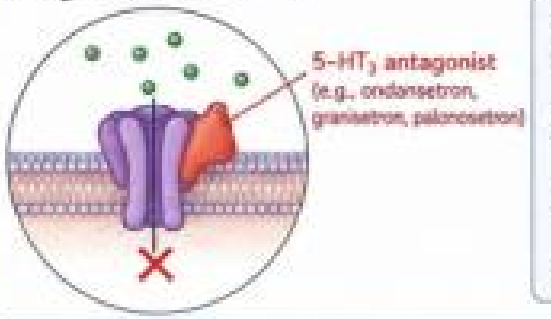
3. SIGNAL TRANSMISSION TO THE BRAIN

Impulses travel via vagal afferents to the nucleus tractus solitarius (NTS) in the medulla → activates the vomiting center.



4. ACTION OF 5-HT$_3$ ANTAGONISTS

5-HT$_3$ antagonists competitively block 5-HT$_3$ receptors on vagal afferent terminals.



- ✖ Prevents serotonin binding
- ✖ Blocks cation influx and depolarization
- ✖ Prevents nerve impulse generation
- ✖ Inhibits activation of vomiting center
- ✓ Prevents nausea and vomiting

KEY POINTS

- 5-HT$_3$ receptors are ligand-gated cation channels located on vagal afferent neurons in the GI tract and in the CNS (area postrema, NTS).
- 5-HT$_3$ antagonists are peripherally and centrally active.
- Most effective for prevention of chemotherapy- and radiotherapy-induced nausea and vomiting.

DRUG CLASS
Selective 5-HT$_3$ receptor antagonists

EXAMPLES
Ondansetron, Granisetron, Tropisetron, Palonosetron, Dolasetron

THERAPEUTIC USES
Prevention and treatment of nausea and vomiting caused by chemotherapy, radiotherapy, and post-operative state.

OVERALL MECHANISM: Blockade of 5-HT$_3$ receptors → Inhibition of vagal afferent signaling → Decreased activation of vomiting center → Antiemetic effect.

Drug	Generation	Half-life	Main Clinical Use	Unique Features	Common Adverse Effects
Ondansetron	First generation	3–5 hours	Chemotherapy-induced nausea & vomiting (CINV), postoperative nausea and vomiting (PONV)	Most commonly used; available oral, IV, ODT	Headache, constipation, QT prolongation
Granisetron	First generation	8–12 hours	CINV and radiotherapy-induced vomiting	Longer duration than ondansetron; transdermal patch available	Headache, constipation
Dolasetron	First generation	7–9 hours	CINV (less commonly used now)	Prodrug converted to hydrodolasetron	Higher risk of QT prolongation
Tropisetron	First generation	7–8 hours	CINV	Also weakly antagonizes nicotinic receptors	Headache, dizziness

Palonosetron	Second generation	~40 hours	Delayed CINV prevention	Highest receptor affinity; prolonged action; superior for delayed emesis	Headache, constipation, less QT prolongation
Ramosetron	Second generation	~9 hours	PONV and CINV	Strong receptor binding with prolonged effect	Constipation, headache

A 58-year-old woman ovarian carcinoma with presents to the outpatient oncology clinic for her second cycle of cisplatin-based chemotherapy. Approximately two hours after the infusion begins, she experiences severe, acute nausea and multiple episodes of vomiting. The oncologist prescribes a medication that selectively blocks a specific receptor type in the central nervous system and the gastrointestinal tract to manage her symptoms.

Which of the following is the most likely mechanism of action or characteristic associated with the drug prescribed?

(A) Actively blocks neurokinin-1 (NK1) receptors in the area postrema
(B) Functions by antagonizing dopamine D2 receptors in the CTZ
(C) May cause a dose-dependent prolongation of the QT interval on an ECG
(D) Primarily exerts its effects by activating cannabinoid receptors
(E) Is highly effective for motion sickness and vestibular-induced emesis

placement due to congenital long QT syndrome is admitted to the hospital for his first cycle of high-dose cyclophosphamide and doxorubicin for diffuse large B-cell lymphoma. To prevent acute chemotherapy-induced nausea and vomiting (CINV), the oncologist plans to initiate a 5HT3 receptor antagonist. Given the patient's underlying cardiac history, the oncologist selects palonosetron instead of ondansetron. Which of the following properties best explains why palonosetron is preferred over ondansetron in this patient?

(A)It lacks inhibitory activity on the delayed rectifier cardiac potassium current
(B)It primarily exerts its antiemetic effect via central neurokinin-1 (NK1) receptor antagonism
(C) It has a significantly shorter plasma half-life, minimizing total drug exposure
(D) It serves as a competitive antagonist with a lower binding affinity for the 5HT3
(E) It rapid renal clearance without undergoing hepatic cytochrome P450 metabolism

NEUROLEPTICS AS ANTIEMETICS

Work mainly by D_2 receptor antagonism in the chemoreceptor trigger zone (CTZ)

MECHANISM OF ACTION

Neuroleptics (dopamine D_2 receptor antagonists) block D_2 receptors in the CTZ in the area postrema and other emetic pathways → inhibit emetic signals → prevent nausea and vomiting.



EMETIC PATHWAYS

CTZ (Area postrema)
- Dopamine
- Chemotherapy
- Drugs
- Metabolic toxins

Vomiting center (Nucleus tractus solitarius)

NEUROLEPTICS
Block D_2 receptors in CTZ

Nausea and Vomiting

APREPITANT – MECHANISM OF ACTION

Aprepitant is a selective Neurokinin-1 (NK_1) receptor antagonist

1. BACKGROUND

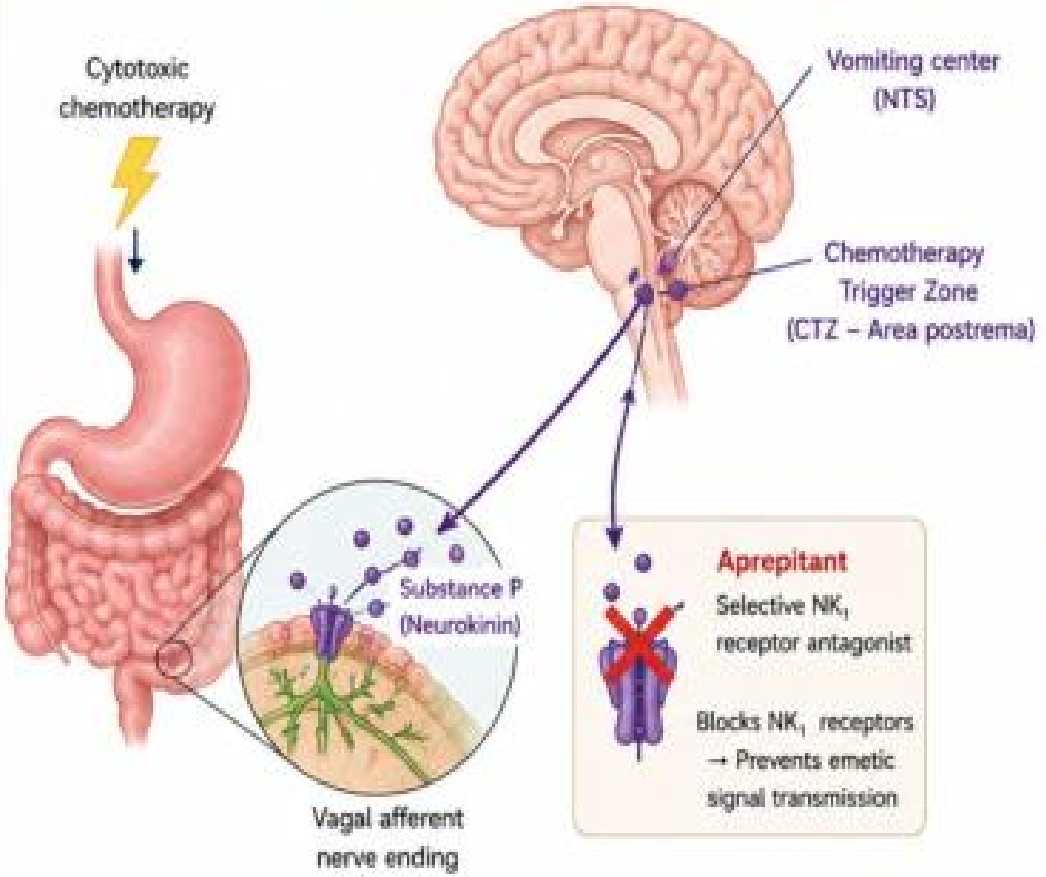
Chemotherapy and other emetogenic stimuli cause release of Substance P (a neurokinin) from vagal afferent nerve endings in the gastrointestinal (GI) tract.

Substance P binds to NK_1 receptors in the nucleus tractus solitarius (NTS) in the medulla (vomiting center) and in the chemotherapy trigger zone (CTZ) in the area postrema → initiation of nausea and vomiting.

2. MECHANISM OF ACTION

- Aprepitant competitively and selectively blocks NK_1 receptors.
- Prevents binding of Substance P.
- Inhibits transmission of emetic signals to the vomiting center.
- Effective for both acute and delayed phases of chemotherapy-induced nausea and vomiting (CINV).

EMETIC PATHWAY AND SITE OF ACTION



3. RESULT

- Inhibits activation of vomiting center and CTZ
- Prevents nausea and vomiting induced by chemotherapy
- Effective for delayed emesis (>24 hours)

4. CLINICAL USES

- Prevention of acute and delayed CINV with highly and moderately emetogenic chemotherapy
- Used in combination with a 5-HT_3 antagonist (e.g., ondansetron/palonosetron) and dexamethasone for optimal efficacy

SUMMARY

Aprepitant blocks NK_1 receptors, preventing Substance P from activating the vomiting center (NTS) and CTZ in the brain → reduces both acute and delayed chemotherapy-induced nausea and vomiting.

A 63-year-old woman undergoing cisplatin chemotherapy develops severe nausea and vomiting 3 days after treatment. She is given a drug that blocks Substance P from binding in the area postrema and nucleus tractus solitarius.

Which drug was most likely administered?

A. Ondansetron
B. Metoclopramide
C. Aprepitant
D. Scopolamine
E. Meclizine

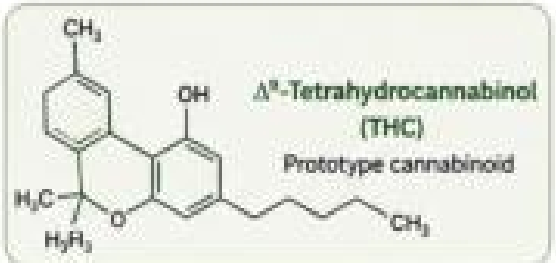
CANNABINOIDS AS ANTIEMETICS

Cannabinoids reduce nausea and vomiting by acting on CB_1 receptors in the brain and GI tract

1. MECHANISM OF ACTION

Cannabinoids (e.g., dronabinol, nabilone) are agonists at cannabinoid CB_1 receptors.

- CB_1 receptors are located in the chemoreceptor trigger zone (CTZ) in the area postrema, the vomiting center (NTS) and throughout the GI tract.
- Activation of CB_1 receptors inhibits release of emetogenic neurotransmitters and reduces neuronal excitability.
- Result: Inhibition of vomiting reflex → antiemetic effect.

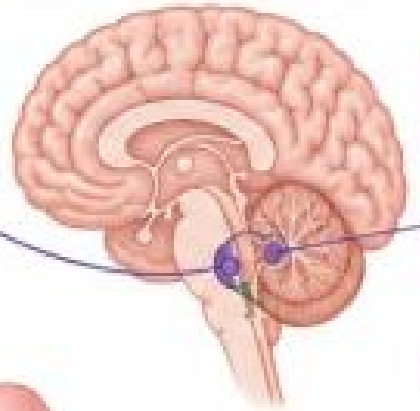


Δ^9-Tetrahydrocannabinol (THC)
Prototype cannabinoid

2. SITES OF ACTION

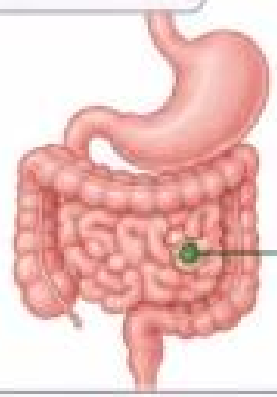
Chemoreceptor Trigger Zone (CTZ) Area postrema

CB_1 activation ↓ Inhibits emetic signal initiation



Vomiting Center (Nucleus Tractus Solitarius, NTS)

CB_1 activation ↓ Inhibits vomiting reflex



GI Tract (Enteric Nervous System)

CB_1 activation ↓ Decreases GI motility and secretion

- Central sites (CTZ, NTS)
- Peripheral sites (GI tract)

3. CLINICAL USES

- Chemotherapy-induced nausea and vomiting (especially refractory cases)
- Appetite stimulation in AIDS and cachexia
- Nausea and vomiting in palliative care
- Gastroparesis (adjunct)



Common Drugs

- Dronabinol (synthetic THC)
- Nabilone (synthetic cannabinoid)
- Nabiximols (THC:CBD oromucosal spray)

4. KEY POINTS

- CB_1 receptor agonists.
- Act both centrally (CTZ, NTS) and peripherally (GI tract).
- Effective for acute and delayed nausea and vomiting.
- Useful in refractory cases not controlled by other antiemetics.



CB_1 receptor activation

Inhibits release of Substance P, 5-HT, Dopamine and other emetogenic mediators

CB_1 receptor

Post-synaptic neuron

5. ADVERSE EFFECTS



CNS: Drowsiness, dizziness, euphoria



Psychiatric: Anxiety, paranoia, hallucinations



CVS: Tachycardia, postural hypotension



Others: Dry mouth, impaired memory, red conjunctiva

6. CONTRAINDICATIONS / PRECAUTIONS

- Pregnancy and lactation
- Severe psychiatric illness
- Caution in cardiovascular disease
- Avoid with other CNS depressants (additive sedative effects)

	MAJOR USES	COMMON ADVERSE EFFECTS	SPECIAL POINTS
PROCHLORPERAZINE	• Nausea & vomiting of various causes • Severe vertigo (labyrinthine) • Migraine-associated vomiting	• Drowsiness, dizziness • Hypotension • Extrapyramidal symptoms • Anticholinergic effects	Potent antiemetic; causes sedation.
CHLORPROMAZINE	• Refractory nausea & vomiting • Hiccups	• Sedation • Hypotension • Anticholinergic effects	More sedating; use when sedation is desired.
PROMETHAZINE	• Nausea & vomiting • Motion sickness • Postoperative nausea	• Drowsiness • Anticholinergic effects • Hypotension	Also H_1 blocker – useful in allergy & motion sickness.
TRIFLUOPERAZINE	• Nausea & vomiting • Intractable hiccups	• Extrapyramidal symptoms • Sedation • Hypotension	Less sedative than chlorpromazine.
HALOPERIDOL	• Refractory nausea & vomiting • Palliative care • Migraine-associated vomiting	• Extrapyramidal symptoms • Akathisia • QT prolongation	High potency; effective in low dose.
DROPERIDOL	• Postoperative nausea & vomiting (PONV) • Refractory vomiting	• Sedation • Hypotension • QT prolongation	Very potent; rapid onset.

SUMMARY

Drug/Class	Mechanism	Major Use	Major Toxicity
5-HT_3 antagonists	Block serotonin receptors	CINV, PONV	QT prolongation
Metoclopramide	D_2 block + 5-HT_4 agonist	Gastroparesis	EPS, hyperprolactinemia
NK_1 antagonists	Block Substance P	Delayed CINV	Fatigue
H_1 blockers	Vestibular suppression	Motion sickness	Sedation
Scopolamine	M_1 blockade	Motion sickness	Dry mouth
Cannabinoids	CB_1 agonists	Refractory CINV	Euphoria

THANK YOU